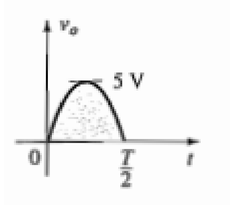
Assignment 3 Answer Key

1. $V_0 = 12\,V$, LED will turn ON.

2. $V_0 = 0.7\,V$, AND Gate

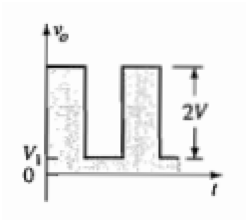
3.



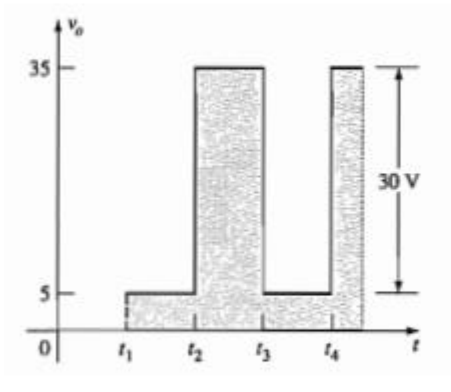
4. $23.76 \leq V_i \leq 36.87\ Volts$

5. $V_{01} = 4\,V$, $V_{02} = 10\,V$, $I_R = 20\,mA$, $P_s = 800\,mW$, $P_{LED} = 80\,mW$, $P_Z = 120\,mW$

6.



7.



8. $V_0 = -(10V_1 + 33V_2)$ Volts = -[0.5 sin(1000t) + 0.33 sin(3000t)] Volts

9. $V_0 = 0.729\ V$

10. -3.8 V, -1.425 mA.

11. 0.8 μA to 1 μA

12. 7 Volt

13. $V_0 = -20(V_2 - V_1)$

14. $V_0 = 21(V_1 - V_2)$